

AI DEFENSE MECHANISM FOR PHISHING AND MALWARE ATTACKS –
SAFEGUARDING INBOX

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Research Proposal

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LIST OF ABBREVIATIONS

AI	-----	Artificial Intelligence
HTML	-----	Hyper Text Markup Language
IP	-----	Internet Protocol
BERT	-----	Bidirectional Encoder Representations from Transformers
XGBoost	-----	eXtreme Gradient Boosting
GPU	-----	Graphics Processing Unit
AWS	-----	Amazon Web Services
CUDA	-----	Compute Unified Device Architecture
RAM	-----	Random Access Memory
GB	-----	Gigabyte
AMD	-----	Advanced Micro Devices
SSD	-----	Solid State Drive

ABSTRACT

Across the globe, we keep hearing about organisations falling prey to phishing and malware attacks, resulting in their collapse. These attacks have become very precise and accurate in today's time. Regardless of the implemented email spam filtering tools or advanced security firewalls installed in systems, malware, spyware, and viruses find a way in, breaching the security. Being a first-hand witness of the scenario in my workplace, I have observed several colleagues of mine, fall for phishing mails that got blended seamlessly into our daily communication. These situations have proven that there is a need of smarter and supportive security tools. We are all aware that phishing attacks have evolved in recent times due to modern technological advancements. Therefore, the limitation of the traditional detection systems also needs to widen. The research will thereby focus on establishing an AI-based framework that supports safer communication environments, by primarily aiming to strengthen inbox security. The expected outcome is to reduce the burden on end-users and address the evolving nature of email threats which could be harmful and result in financial losses in various organisations, institutions and businesses.

1. Background

Electronic mails continue to be widely used communication channels. They are utilized by both professional and non-professional users across the globe. This also makes it dominant entry gateways for cyberattacks and malicious hacks [1]. With new technological advancements, the phishing attacks have become much more sophisticated. This has led to significant security breaches and impacted financial losses in multi-millions [2]. Such threats have caused downfall of major companies and organisations and have impacted employees of the workplace, both emotionally and financially. Once such threats find their way into the systems, the security deems vulnerable, if not curated adequately to identify the threat. The phishing hacks and malware attacks, are designed to bypass the installed security systems and tools. These attacks use subtle social engineering techniques and depend on human errors to gain access to systems. It is evident that business operations have got impacted at a large-scale followed by loss in revenue. The strike rate of attacks is more than 95% [3]. This makes the problem much bigger and more urgent. These malware attacks not only exploit technical vulnerabilities and loop holes in systems but also breaks a user's trust.

All the organizations, whether small scale, large scale or be it multi-national companies are relying on email for ease of communication. It is hence, extremely challenging to differentiate between authentic legitimate mails and malicious mails. Increased user base and digital footprint have also added to this cause of systems being more attack prone, leading to high exposure to attackers and hackers.

To protect from such threats and attacks, organisations have deployed security measures such as basic anomaly checks, certain rule-based filtering, detecting incorrect sender's mail address, static signature detection, etc [4]. However, these methods are also not completely reliable and have become outdated to a large extent. Modern day attackers have learned and are very successful in bypassing the traditional security tools. Their attack strategies have now evolved. The phishing attack is being carried out using deceptive language and certain language-based cues. The mail content though subtly but very convincingly impersonates. The root cause observed is that the conventional security systems were not built to interpret the intention or read between the lines by capturing anomaly patterns. They were built using limited rigid rules of detection and identification and thereby fail to recognise the attacks.

At my workplace, I have witnessed, several colleagues falling prey to phishing emails. These communications often look legitimate and disguise as internal communications. The mail communications are so well-crafted that they look like routine mails. Even well-informed employees have taken rushed actions out of fear and misjudgements when they received urgent alerts and warning mails. They have clicked on malicious links and downloaded harmful attachments. This has resulted in security incidents impacting the job of the respective employee. Such user actions lead to credentials getting compromised. It also leads to an act of data breach of the organisation. The information security department have stepped in to resolve such cases, in order to prevent further damage. However, at times, the damage might have already occurred in due time. There could be momentary lapse in judgements by the user due to various factors such as workload and strenuous hours.

The research is thereby focussed to overcome the shortcomings of the current email security mechanism. The need of artificial intelligence driven solutions is being sought to design a multi-layered framework integrating, generative AI, python, natural language processing and machine learning techniques [5]. An integrated approach of these advanced learning technologies will aim at strengthening inbox security. The idea is also to explore ways to train the AI framework to become resilient against evolving threats and provided real-time protection.

2. Literature Review

This section emphasizes on various studies and researches performed on email security and threats, detection methods – both traditional and modern, machine learning based solutions, deep learning-based solutions, and AI-driven solutions. Some studies have shown how the threats have evolved with advancement in technology, leading to the traditional detection methods proven ineffective. With more subtle and sophisticated methods, attacks are more specific. The review will also highlight the gaps discovered in other studies.

- a) A study sourcing Cornell University data shows that when 2000 phishing mail samples were analysed, it was observed that the phishing trends have changed over the period of time [\[11\]](#). The patterns are now more sophisticated and text-based mistakes are minimal to zero.
- b) Email communications continue to be dominant entry points for phishing and malware attacks. But, human errors and lack of judgement are also vulnerable behaviours, through which the users are deceived [\[12\]](#). The messages are target-specific and to trick the receivers of the mails, often personal details and information are used. The success factor of a phishing or malware attack is one human error.
- c) To overcome known threats, multiple studies have been carried and to a larger extent have been also implemented in real-time environments. Standalone techniques of digital signatures to verify authenticity, hashing algorithms and encryptions are in place protect the email environment [\[13\]](#) as first line of defense. But, with attacks evolving much smarter, these techniques are likely ineffective and will struggle to protect the users against unseen threats.
- d) In cybersecurity, machine learning [\[14\]](#) and natural language processing techniques have overcome the barrier of rigidity. Supervised and unsupervised learning methods, Use of various machine learning models to learn which approach works best in tackling the patterns of threat and spams. But these model, may lack in capturing certain adversities that can evade the machine learning features. Unless the models are trained to actively learn, novel attacks cannot be prevented. The machine learning models have also limitations in handling linguistic tricks and manipulation. They might fail to identify and continuously adapt to the changing behaviour and intent of the mail content.
- e) On the other hand, studies on ensemble models [\[15\]](#) have shown adaptability to changing patterns and also, to an extent are scalable. The accuracy and performance in hybrid models are better than traditional machine learning. The reason is their ability to

deduce a deeper level of semantics understanding, how words are related. They are scaled to process context.

Gaps seen in existing research: Earlier studies focussed on exploring single detection methods. This later was upscaled to combine with machine learning, natural language processing techniques. A few studies have also explored a combination of the above mentioned with generative AI, which showed enhanced performance. However, model training is reliable on older datasets. There is also a gap between practical application, which can be built to be implemented in real-time and designing a theory. Variations can be observed in practical implementation compared to the findings of the theoretical study.

This research aims to bridge these gaps by using several AI techniques, leveraging the strength of an ensemble model to build a multi-layer defense mechanism. It is an attempt to assist other existing studies and also to handle the gaps in future time. The approach differs in the below mentioned aspects:

- It is bringing together multiple signals within one framework.
- For testing the models, richer and varied datasets are required. These are synthetically yielded through Generative AI.
- The layered learning method will give the system a certain edge, wherein the models work together and not behave individually.
- Zero-day threat testing to evaluate unseen threats.

3. Research Questions

The study of the research is guided by the following questions:

- 1) How can artificial intelligence methods help the email systems to identify malicious and suspicious mails and messages in a much more effective manner?
- 2) How artificial intelligence can further enhance the existing security measures?
- 3) What aspects of an email, such as words, sentence formations, grammar mistakes, sender's mail address, or any other hidden metadata can prove to be more evident in spotting the phishing mails from genuine mails?
- 4) In what ways can the natural language processing and machine learning techniques be utilized to understand the intent of any email being sent to a receiver?
- 5) Does a combined approach of both Natural Language Processing and machine learning and techniques, prove better in detecting and catching hidden malwares within any email content such as scanning of attachments which could contain a malicious virus?
- 6) Can a combination of AI models be trained to perform a deeper email analysis and detect the underlying threat in an email?
- 7) Is the application of multi-layered defense strategy feasible in real-world environments?
- 8) How well the proposed AI framework function if exposed to zero-day attacks? Can it also eradicate the vulnerabilities missed in previous mail threats, which passed unseen?

4. Aim and Objectives

Aim: To design a multi-layered AI driven framework that safeguards email inboxes by robust detection and identification of phishing and malware attacks.

Objectives: The primary objectives of the study are:

- To curate and implement a layered defense system or tool by using artificial intelligence, machine learning and natural language processing methods, in order to identify and flag malicious mail content.
- To make the framework capable of detecting warnings and negative indicators of any suspicious activity, hidden in the mail content in the form of attachments, embedded links or sender's mail address or details.
- Evaluation of different models and testing the implementation using various methods such as natural language processing classifiers, deep learning networks and anomaly detection algorithms, to verify how well they function together.
- To also measure the performance of AI system against the conventional detection systems in place.
- To measure the effectiveness of AI framework when it faces zero-day scenarios.
- To identify the real time challenges being faced by organisations in implementing and deploying AI enhanced security to their inboxes.
- To strive to contribute if inboxes can be made more resilient with evolving threats.

5. Significance of the Study

The study is important as it is addressing an urgent problem of unsafe email communications in both professional and non-professional sections of the population. By building an intelligent email threat detection system, the gaps observed in rigid, rule-based detection systems can be overcome. Implementation of multi-layered AI defense framework which is capable of adapting dynamically and can be further trained to build resilience will safeguard users and organisations in the longer run.

6. Scope of the Study

The scope of the study solely focusses on designing and testing a multi-layered AI defense framework aimed at email security. The study revolves around identifying threats by analysing who is the sender, what is the content of the mail and how is it written, what type of attachments or links are sent, what is the content of the attachments and links. By studying these patterns and behaviours, the framework is aimed to be made smarter and dynamically adaptive towards evolving threats. Further, to add, the implication of deploying the solution in real organization will be discussed. But the implementation is restricted towards any organizational and/or corporate environments.

The study does not address physical security concerns or issues, network intrusion detection [\[6\]](#), etc. The tools needed to study the mentioned issues is eminent but is out of scope for this respective study.

7. Research Methodology

To design and implement the AI-driven email defense framework, Generative AI, Machine Learning, Natural Language Processing and Python language are used. Python serves as the primary development environment. It has broad ecosystem of AI libraries.

The research design will involve the 5 phases:

- Gathering and Preparation of real and computer-generated (synthetic) email datasets.
- Using Generative Artificial Intelligence models [7] to understand the hidden semantic attributes.
- Training of machine learning models by feeding processed data in machine learning classifiers.
- Build a multi-layered architecture using Python that combines different AI models.
- Evaluation and comparison of the performance of system against the existing baselines.

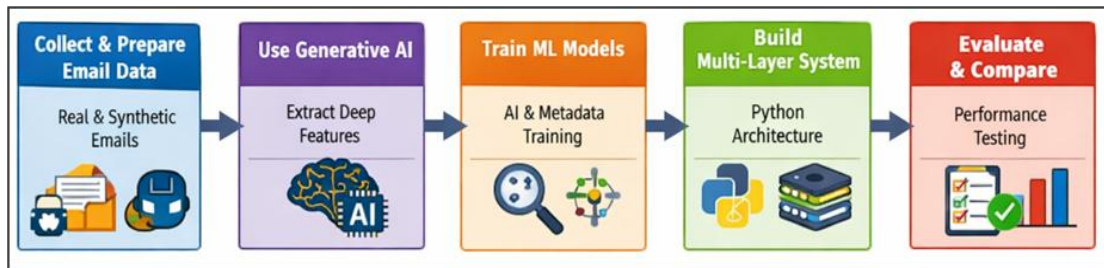


Figure 1: Phases of Research Design

a) **Phase 1- Data Gathering, Classification and Pre-processing:** Collecting and creating a mix of real and artificially created datasets. Real email datasets gathered from publicly available datasets such as PhishTank, PhishingCorpus, Enron Email Dataset, SpamAssassin Dataset, etc. These data will serve to have a varied collection of metadata, attachments, links, etc. Datasets created using Generative AI serves to be more realistic, anonymized. They can imitate the ambiguous and deceiving email content and patterns. Using Generative AI, impersonation attacks can be simulated. A variety of phishing email templates will prove rich dataset for testing phase. The data attributes focus is on: Email body text, Email Subject, Metadata of sender (IP address, domain, geolocation) and embedded attachments and links.

Once datasets are collected, classifying of the emails is required. This is done under four categories:

- Safe
- Threat/Malicious
- Phishing

- Malware

The email datasets need to be cleaned and prepared for further usage. Python libraries such as NumPy and pandas will be utilized. The pre-processing steps include:

- Removing unnecessary parts like redundant HTML tags, irrelevant scripts in order to achieve text cleanup.
- Normalizing the text into standard format by tokenization (splitting text into words or phrases)
- Identifying and removing stop words such as 'the', 'is', 'and' etc. These words do not add any efficiency while detection.
- Fetching of useful metadata such as the domain of the sender, pattern of links, routing pathways. Suspicious and malicious are indicated through this data.
- Lastly, the text and metadata (non-numeric data) are to be converted into numeric formats. This process is called 'Feature encoding'. It enables machine learning models and AI models to process information effectively. It also improves performance of the model.

Once these steps are carried out, the cleaned and processed data is fed into the machine learning models for training and further evaluation.

b) Phase 2- Feature Engineering using Generative AI: The embedded content within emails can be read to detect hidden patterns. These are often missed in traditional rule-based filtering mechanisms. The generative AI models such as Sentence Transformers, which is a Python framework and library based on transformer models like Bidirectional Encoder Representations from Transformers (BERT) assist to understand the email texts in depth and analyse the meaning [\[8\]](#). They help in capturing:

- Meaning of words and sentences in use.
- Tone of the message, style of writing.
- Intention of the message by analysing certain cues likely, hinting towards urgency or any sort of imitation or mimicking attribute.

In addition to the embeddings, the system will be capable to understanding behavioural and grammatical signs in the email content. This combination supports more sophisticated malicious intention in the email. These signs are basically clues of identification and contain:

- If someone is impersonating anybody else's identity in order to appear trusted to the receiver of the email.

- Patterns that show urgency by the sender expecting a quick action or response from the receiver of the email.
- Usage of words and sentences very similar to those in phishing mail content and templates.
- Validating the emotion and the tone of the email message. It could sound urgent, friendly, warning-based or could appear to be threatening.

c) Phase 3- Development of Machine Learning Models: The study will use Python language along with some well-known AI libraries, namely, TensorFlow, PyTorch, XGBoost and scikit-learn. These will be used to build and train models to detect malware and suspicious content in emails. The AI framework will use the below listed models:

- Logistic Regression: This is a simple method used for classification. Based on data patterns, this model helps classify emails as safe, threat, spam, etc [\[10\]](#).
- XGBoost: This is a strong boosting algorithm that focusses on minimising errors and improving the accuracy. It learns from past errors and mistakes by iterative refinements [\[9\]](#).
- Neural Network Classifiers: This is a supervised learning model. It is important in the application of training models. It is essential for analysis and understanding the content of the mail, including the language used. Through this analysis method, phishing cues can be spotted. Additionally, it also aids in elaborating the understanding of the relationship between the metadata and text.
- Random Forest: This is a machine learning algorithm that uses several decision trees to make enhanced predictions. It helps in arriving at a single output, by also largely reducing errors [\[10\]](#).
- Anomaly Detection: This is an analytical technique, also known as outlier detection. This technique is known to identify and spot observations which are different than the usual data. The analysis seeks for unusual patterns that refrain from matching the normal, usual patterns. It is significant to catch new threats, that might have been missed in rigid detection rule-based filtering techniques.

All these models together are used to create a multi-layer defense system. Each layer has its own significance. This approach helps the system to maintain high accuracy of detecting malware and phishing content, by also simultaneously adapting to evolving threats.

After training the models, the next step is to ensure that the final prediction is accurate and has a strong outcome. This can be achieved by combining the output of each model. The process is known as ‘Ensemble Integration’. This process assists in reducing errors and

eliminate biases. In order to conform, the system only uses the strong parts of each model.

The Ensemble integration process uses three main techniques:

- Voting: In here, each model yields its predictions. The final output is decided by either selecting a hard voting i.e., by majority vote or a soft voting i.e., by average of confidence scores.
- Weighted Averaging: Some predictions are given more weightage during output calculation models, as they may perform better than other models.
- Stacking: This technique stacks the results derived from all the models to arrive at a final output decision with accuracy.

The system becomes more reliable, accuracy is improved and is capable of handling varied types of emails, both suspicious and legitimate ones, with the amalgamation of above the approaches.

d) Phase 4- Building a Multi-layered Python Architecture: The framework will be designed as a pipeline in Python in a stepwise manner. Each layer will have its significance and helps build a system which is more robust and precise in detecting and flagging phishing and malware attacks from various perspectives. Each layer in use adds relevant protection levels in the framework. Listed below is the description of each layer:

- Validation of Metadata: This layer checks the email headers, domain of the sender and routing pathways. It protects from obvious fake “from” addresses, can be called as spoofing.
- Analysis of Uniform Resource Locators and Attachments: This layer scans the email and examines for links and attachments in the mail for any malicious content. It protects the user from unusual websites, phishing links, or virus laden attachments before the mail is opened to be viewed or read.
- Semantic Analysis using Generative AI: This layer makes use of generative AI, to understand the language, meaning, tone and also the intent the mail. It demonstrates spotting the tricks such as presence of urgency words – “act now”, “access blocked”. It can also spot suspiciously written mails and impersonation – mails addressing “familiar business name”, “sender who could be a colleague” or even “a superior at workplace”.
- Classification based on machine learning: This layer applies machine learning models, which are trained on datasets. It classifies emails into categories of safe, spam, threats, etc, entirely based on the learned patterns. This leads to accuracy improvement.
- Detection of Anomalies: This layer looks for unusual and rare behaviours, to help catch new or unseen attacks, in comparison to the normal emails encountered. It will be

trained to prohibit zero-day threats, that are different from existing phishing attack templates.

- Decision of Final Integration: This layer consolidates the best output extracted from all the layers, by using the ensemble integration approaches. Hence, providing the final outcome to reflect unbiased, no errors, and a balanced and accurate judgement.

e) Phase 5- Evaluation of the model performance: After training the models, the next step is to test the performance of the models. This step will determine how well they actually work in detecting the phishing and malware-based emails accurately and consistently. The evaluation shall include:

- Precision: This metric indicates the count of the flagged emails as threats or malware are absolutely harmful.
- Accuracy: This metric indicates the number of times the system is able to correctly identifies legitimate versus malicious emails.
- Recall: This metric indicates how many of the actual positive threats were caught by the system, with zero misses.
- F1-Score: This metric is indicative of measuring balance by combining precision and recall. It is mainly useful when datasets have imbalanced classes. A higher F1 score signifies high performance.
- False Positives and False Negatives: These are error metrics. False Positive indicates if a threat or malware is incorrectly classified as positive i.e., legitimate or genuine. Similarly, False Negative indicates if an authentic mail is flagged as negative i.e., threat or spam email.
- Zero-Day Samples Detection: This is an eminent metric to evaluate. The test confirms how effectively new, unusual or unseen malware samples is handled by the system. These samples will not be a part of the model training. Upon testing the results shall be a measure of the performance.

To make the results simple to understand and visualise, we will use a Confusion Matrix. Through the confusion matrix, it is very easier to view the strong parts of the models and to also deduce, the areas where the model is facing challenge. This observation can further assist in fine tuning the model performance. The Confusion Matrix is represented in four quadrants aligned to the four outcomes. They are:

- True Positive (TP): Threat mails captured correctly.
- True Negative (TN): Safe mails passed correctly.
- False Positive (FP): Safe mails incorrectly flagged as Threat mails.

- False Negative (FN): Threat mails failed to be captured and flagged.

Classes	Predicted Class	
Actual Class	Phishing	Legitimate
Phishing	True Positive (TP)	False Negative (FN)
Legitimate	False Positive (FP)	True Negative (TN)

Figure 2: Confusion Matrix

A comparative study between the existing models which are using conventional rule-based filters and a limited detection design and, the multi-layered AI-driven approach to be carried out. This comparison will show the additional edge that the AI framework is capable of providing in prevention the email inbox of users from phishing and malware attacks.

8. Requirements Resources

For the implementation of the research study, listed below are the resources required.

- **Software:**

- **Programming Language:** Python is the primary programming language.
- **Environment:** Jupyter Notebook or JupyterLab, Visual Studio with Python extension.
- **Key Libraries:**
 - Machine Learning: TensorFlow, PyTorch, scikit-learn, XGBoost
 - Generative AI: SentenceTransformers, HuggingFace Transformers
 - Data Processing: NumPy, pandas
 - Anomaly Detection: PyOD
 - Visualization: PlotLy or Bokeh or matplotlib
- **Version Control:** GitHub as code repository

- **Hardware:**

- Quad-core processor, Intel i5-i7 / AMD Ryzen 7.
- RAM of 8-16 GB
- Storage Capacity of 512 GB – 1TB SSD
- Integrated GPU, NVIDIA GPU with CUDA support (optional)
- AWS Cloud Platform
- Google Colab (optional)

- **Data Sources:**

- Public Phishing Datasets sourced from PhishTank, PhishingCorpus and OpenPhish.
- Email Security Datasets sourced from Enron Email Dataset, SpamAssassin Public Corpus.
- Synthetic Datasets generated using AI
- Augmented Datasets
- Malware Samples from VirusShare, MalwareBazaar.
- Honeypot environment setting with zero-day anomaly data.

9. Research Plan

Provided below is the research plan from initiation to implementation.

Pre-Preparation: Hypothesis formulation [2-3 Weeks]

- Risk:
 - Overload of information sources making it difficult to deduce what is actually relevant.
 - Referencing older studies may impact the review.
 - Broader hypothesis may impact the time of implementation.
- Contingency:
 - Use diversified information but demarcate the inclusions and exclusions clearly.
 - Track latest published papers and studies through reliable sources.
 - Refine the hypothesis based on feasibility.

Phase 1: Data Gathering, Classification and Pre-processing [Month 1]

- Risk: Adequate data set collection of phishing samples might be delayed.
- Contingency: Leverage generating synthetic datasets and even augmenting existing dataset for simulation purpose.

Phase 2: Semantic Analysis [Month 2]

- Risk: There lies a risk of misinterpretation of the language, text and content of the phishing mail samples for study by the semantic models.
- Contingency: To cross-verify and validate with various models. Also, to fine tune for expected outcome.

Phase 3: Model Training [Month 2-3]

- Risk: Datasets might be imbalanced, meaning if too many safe emails are used than phishing mails for simulations, then the model may overfit and could not be able to adapt to unseen threats.
- Contingency: A fair mix of samples to balance the techniques can be implemented. Also, testing can spread to different parts of dataset rather than concentrating on one type of dataset.

Phase 4: Multi-layered Framework Development [Month 4]

- Risk: Common risk of integration failures between layers.
- Contingency: To perform thorough unit testing, before system integration.

Phase 5: Evaluation of Results and Comparative Study [Month 5]

- Risk: The risk of AI trained model not outperforming the rule-based conventional filtering approach.
- Contingency: To re-validate and fine tune the weightage of the amalgamated model and retrain the synthetic datasets to improve accuracy.

Phase 6: Documentation of Findings and Peer review [Month 6]

- Risk: There could be potential delays in documenting the work due to time constraints. Also, peer reviews could be delayed due to unforeseen circumstance.
- Contingency: To document the entire study well and in structured way, it is required to keep a proper record of all the experiments and code versions intact. Plan ahead and keep the peer informed on the progress of the work to gain early feedback to avoid last minute delays.

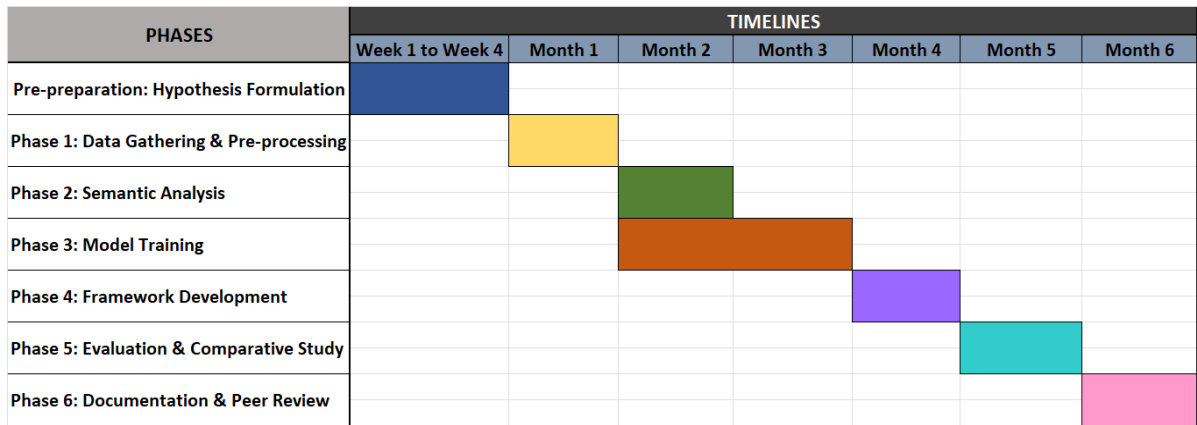


Figure 3: Gantt Chart of Timelines

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